Centroid-Routing and Adaptive MoE for Multimodal Continual Instruction Tuning

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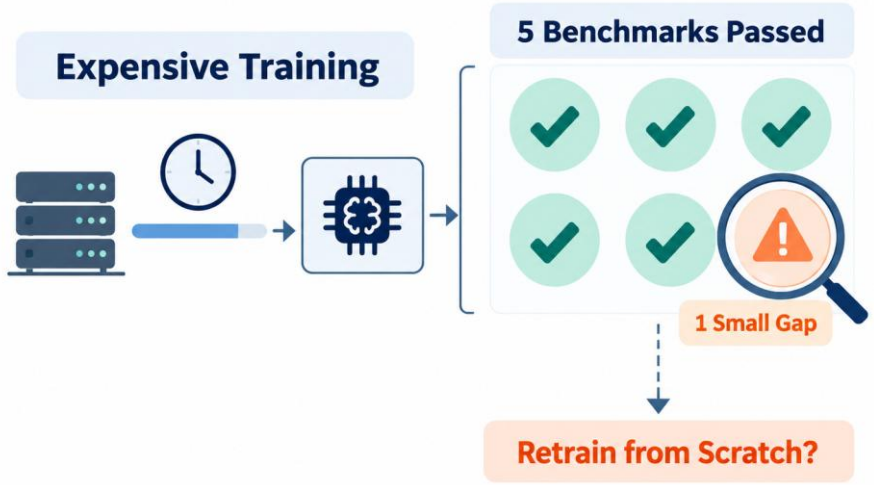
Presenter : Jun-Tao Tang



Background

What problem we try to solve?

1. The model falls short on only a few benchmarks



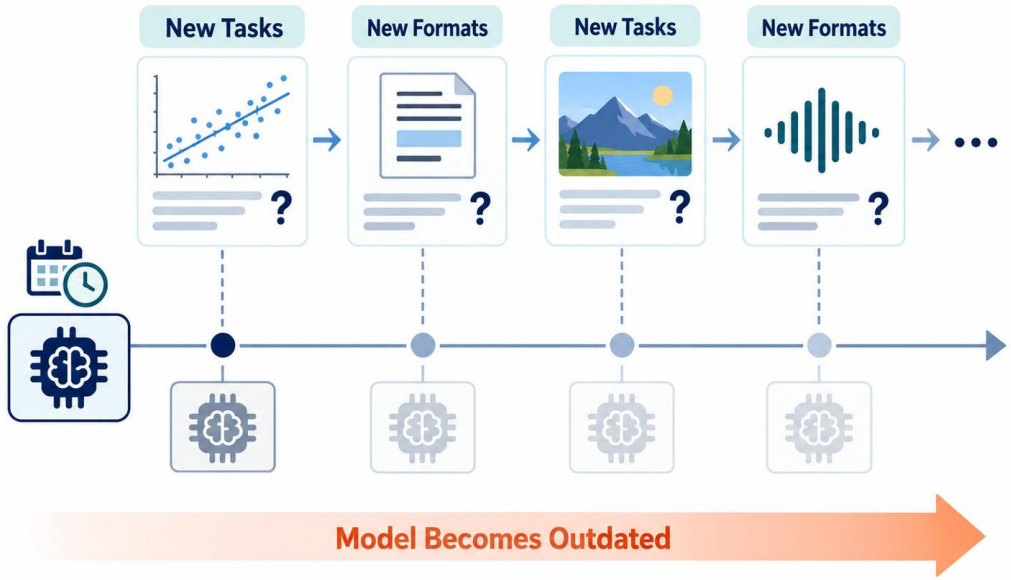
You spend many hours training a model from scratch, only to find that it performs poorly on several benchmarks

- Retrain the model from scratch? Too expensive!
- You might think, “It is only a small gap!”

Background

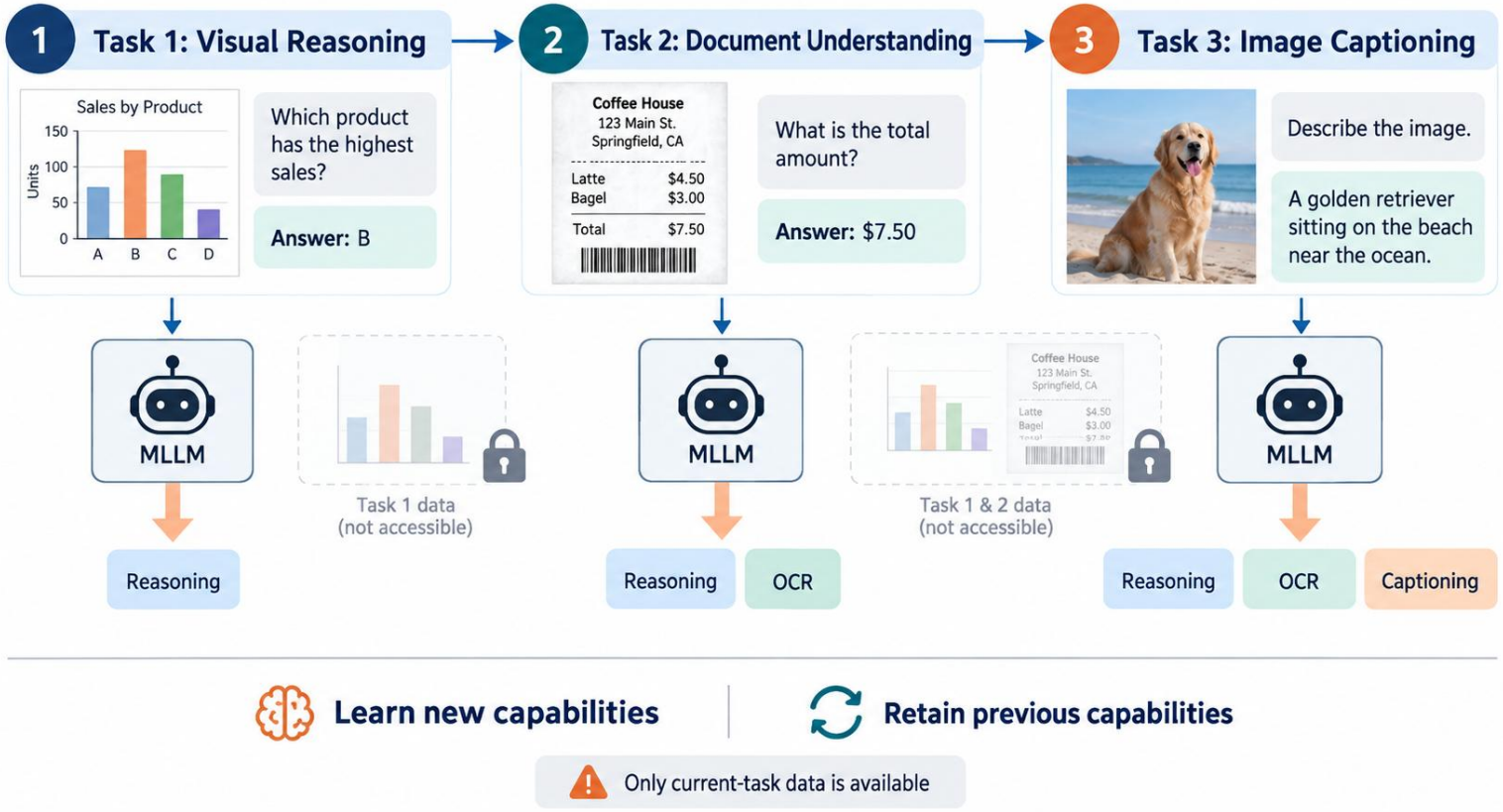
What problem we try to solve?

2. The model is always becoming outdated



- Requirements are constantly changing.
- Leading AI companies, such as OpenAI and Anthropic, must continuously train new models to maintain their competitive edge.

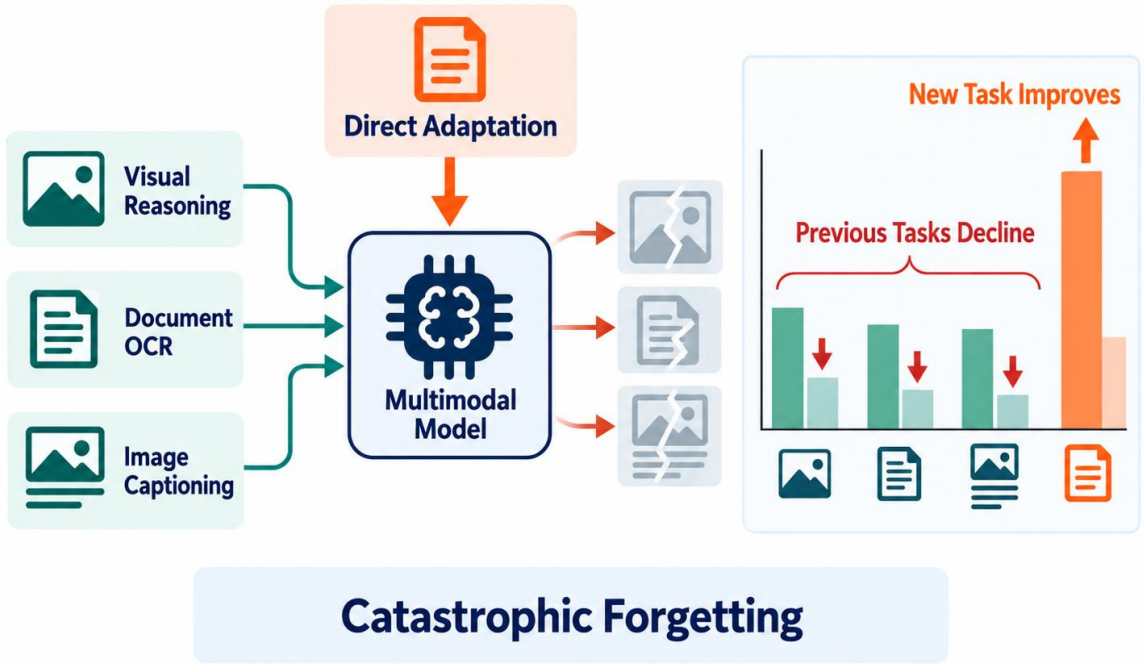
What is Continual Tuning?



Avoid retraining the entire model. Continually expand its capabilities as new tasks arrive.



Problem: Catastrophic Forgetting



1. Learning new tasks may damage existing capabilities

• Adapting the model to a new task can overwrite knowledge learned from previous tasks.

• Performance improves on the new task but declines on earlier tasks.

2. The model should learn without forgetting

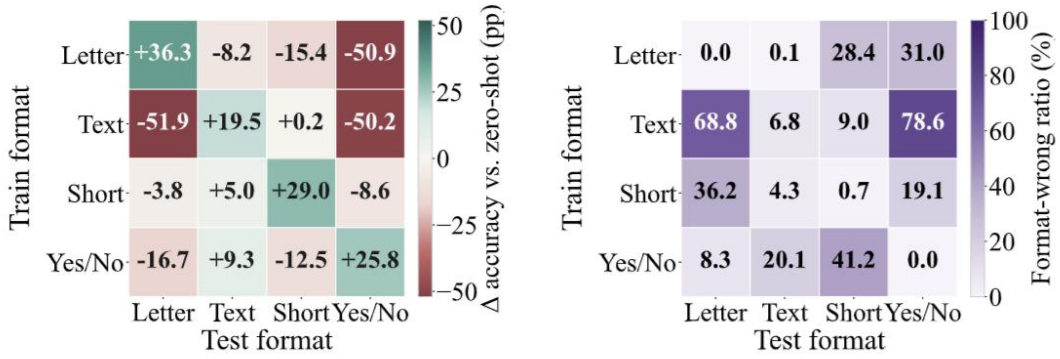
• The key challenge is balancing **plasticity** and **stability**.

Key Findings

How to balance **plasticity** and **stability?**



Finding 1 : What Should Be Isolated?



(a) Change vs. Zero-shot (b) Format-wrong ratio

Figure 1: Format-level interference. (a) Accuracy change relative to zero-shot for each train-test format pair. (b) Ratio of format wrong but semantically correct responses among errors.

Experiment

- We construct four variants of the same dataset using different answer formats.
- We finetune the model on one format and evaluate it on all four formats.

Observation

- Training improves the matched format but often hurts the other formats.
- Many failed responses contain the correct answer in the wrong format.

Finding 2 : What Can Be Reused?

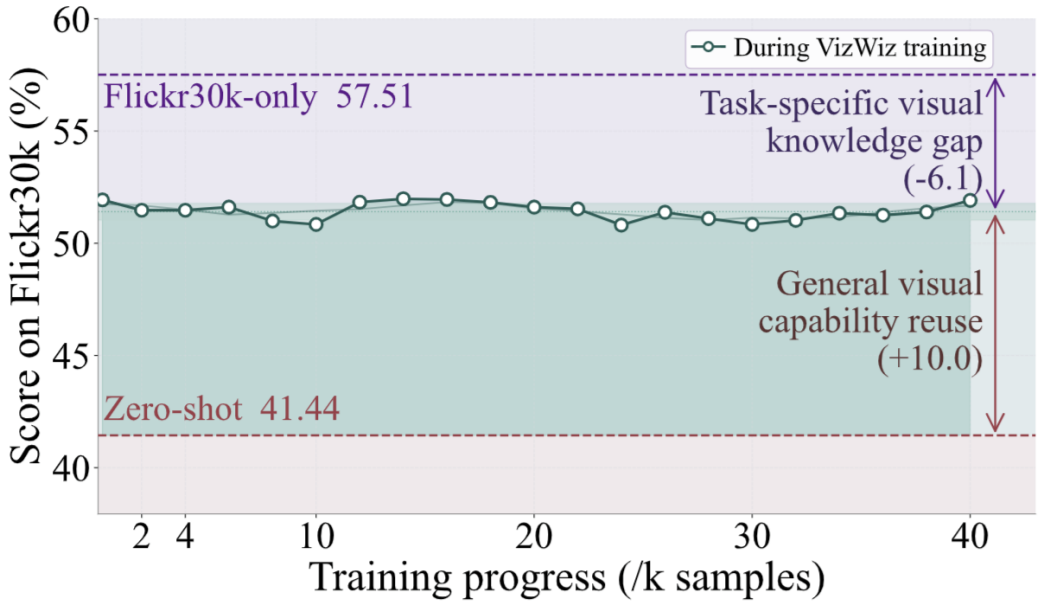


Figure 2: Capability transfer during training.

Experiment

- We train the model on VizWiz and continuously evaluate it on Flickr30k.
- The two datasets use the same answer format but contain different visual domains.

Observation

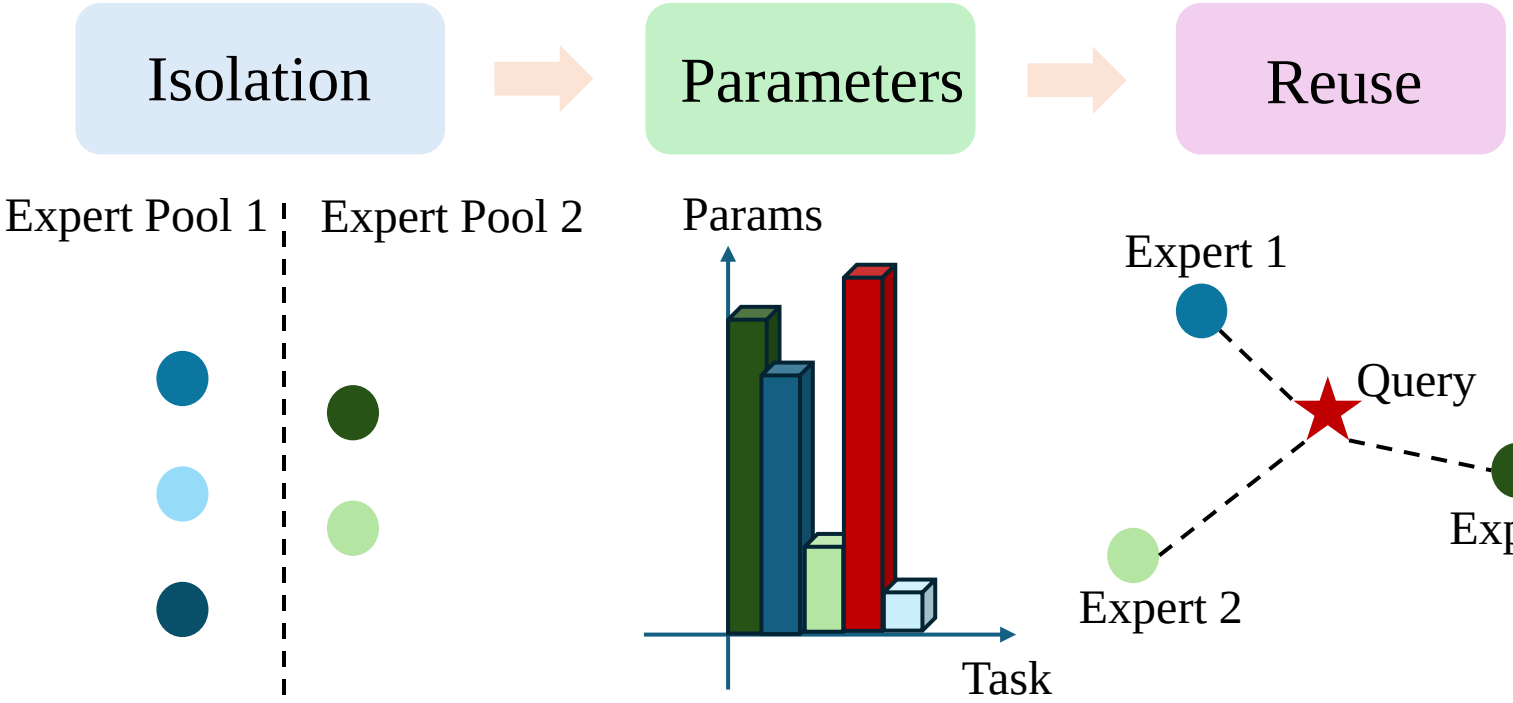
- Training on VizWiz improves Flickr30k performance by 10.0 points, but a 6.1-point task-specific gap remains.

Insight:

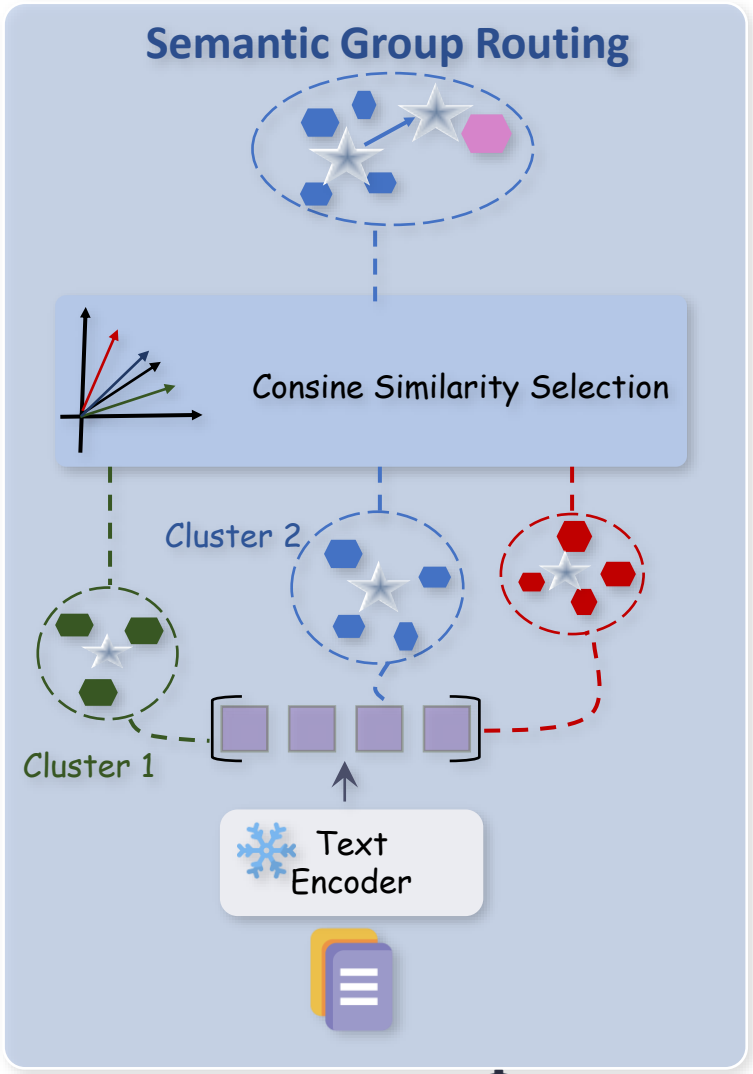
- Output formats are highly task-specific and should be isolated to avoid interference.
- Visual capabilities are partially transferable and should be reused across tasks.



Method


Isolation
Parameters
Reuse
Expert Pool 1
Expert Pool 2
Params
Task
Expert 1
Query
Expert 3
Expert 2

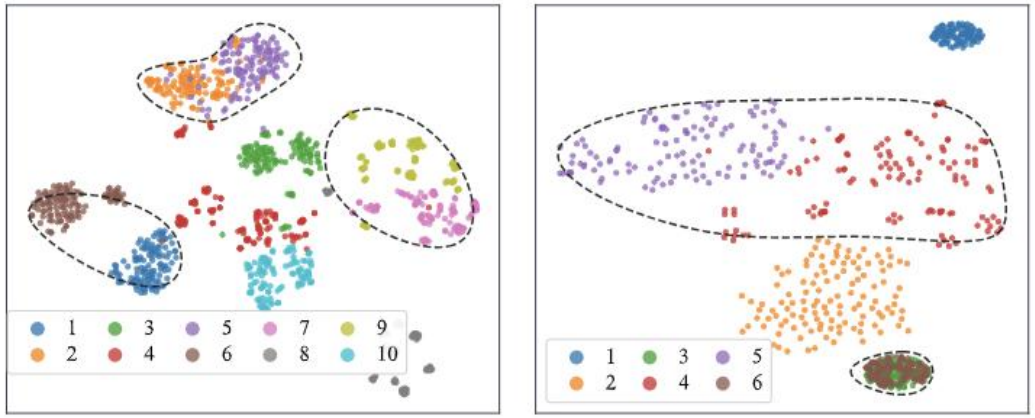
Part 1: Isolate Conflicting Instruction Formats



- Encode each instruction with a frozen CLIP text encoder.
- Assign similar instructions to the same semantic group.
- Create a new group when no existing group provides a confident match.

Part 1: Isolate Conflicting Instruction Formats

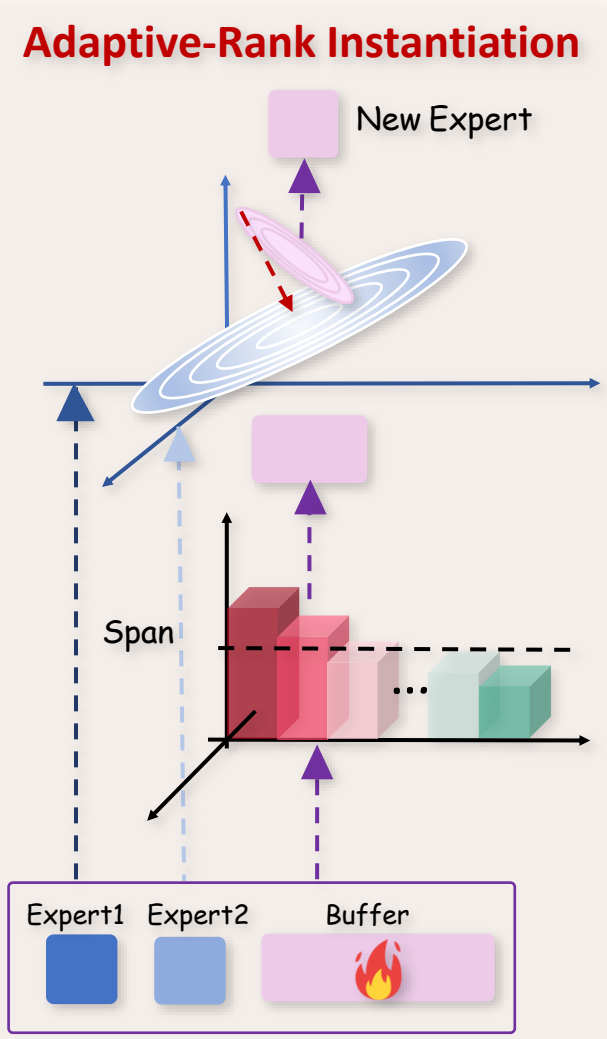
Why can CLIP text encoder reliably group instructions?



(a) TriGap (b) UCIT

Figure 7: Visualization of instruction embeddings on TriGap (a) and UCIT (b). Circled regions indicate instructions assigned to the same group.

Part 2: Allocate Parameters Only for Gaps



How to determine the rank of each LoRA expert?

- A short warm-up estimates the update required by the new task.
- SVD decomposes the update into capability directions.
- Directions already covered by previous experts are filtered out.

$$\alpha_p = \frac{\|(\mathbf{I} - \mathbf{Q}\mathbf{Q}^\top)\mathbf{v}_p\|_2}{\|\mathbf{v}_p\|_2}. \qquad (6)$$

Only directions satisfying $\alpha_p > \tau_{\text{orth}}$ are retained in $\mathcal{S}$ to reconstruct the final expert parameters:

$$\mathbf{A}_t = \mathbf{\Sigma}_{\mathcal{S}}^{1/2}\tilde{\mathbf{V}}_{\mathcal{S}}^\top, \qquad \mathbf{B}_t = \mathbf{U}_{\mathcal{S}}\mathbf{\Sigma}_{\mathcal{S}}^{1/2}, \qquad (7)$$

Part 2: Allocate Parameters Only for Gaps

Different layers and modules require different numbers of parameters.

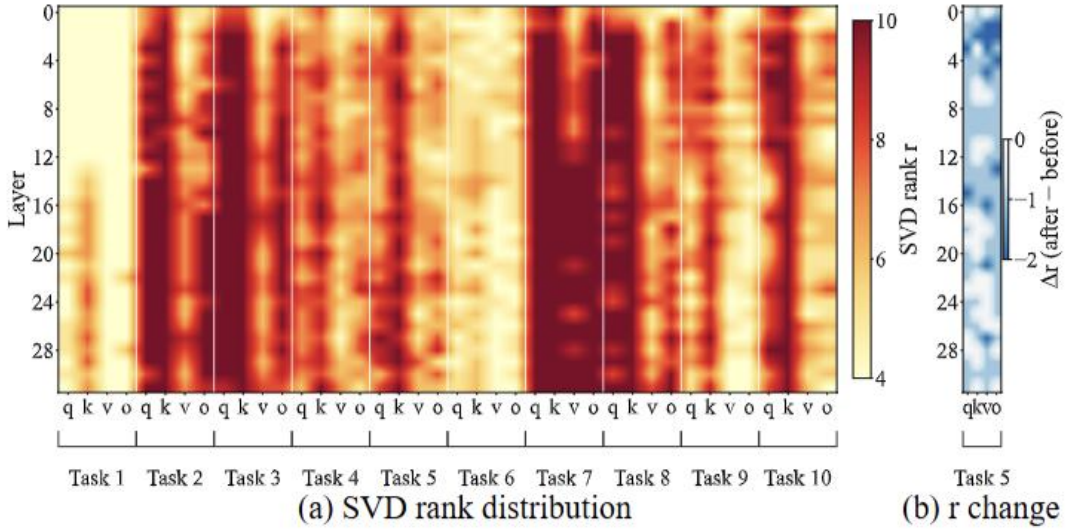
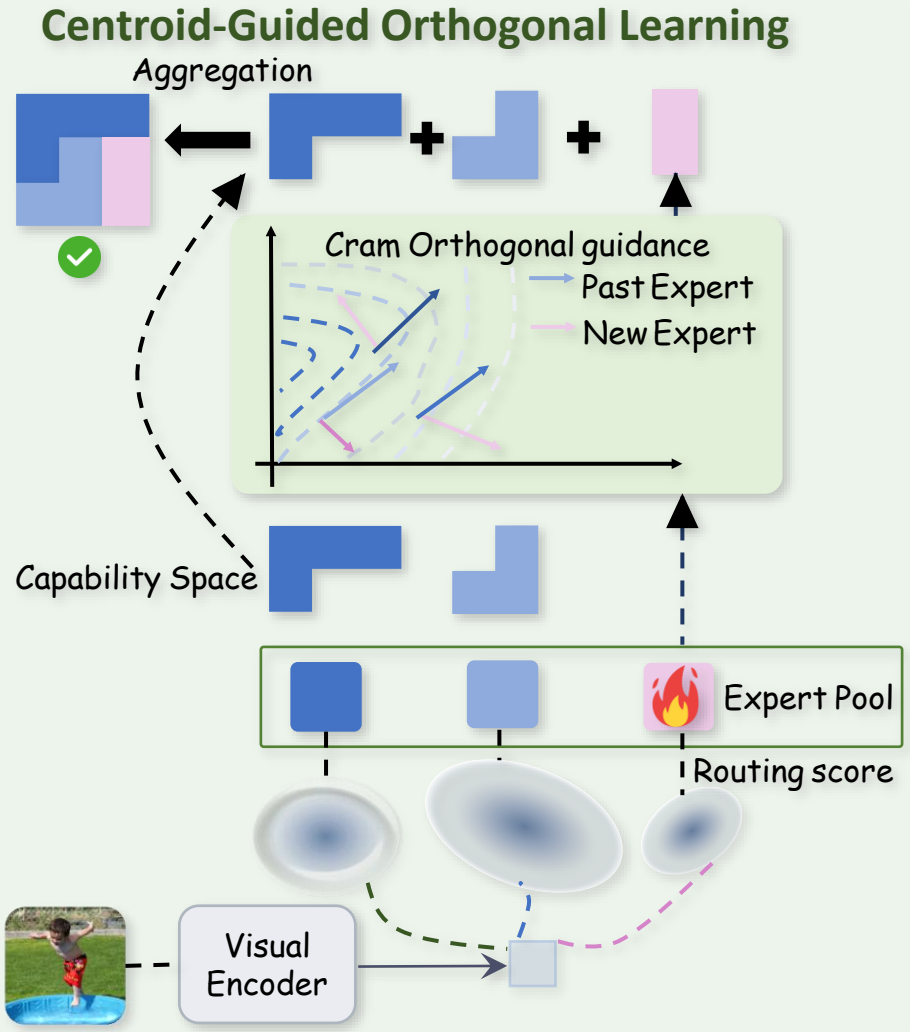


Figure 8: Adaptive rank allocation analysis. (a) Rank distribution across tasks, layers, and modules. (b) Effect of orthogonality filtering: rank allocation for task 5 before/after projecting out historical subspaces.

Part 3: Centroid Routing Enables Stable Reuse



Centroid-Guided Routing

- Each expert has a visual centroid.
- Inputs are routed to the top-k closest experts.
- Previous centroids remain fixed for stable routing.

Orthogonal Learning

- Keep previous experts frozen.
- Train new experts in directions not covered before.

Experiments

- How well can CRAM learn new tasks without forgetting previous ones?
- What is the computational overhead of CRAM?



SOTA Performance

Table 1: Performance on UCIT. The best and second-best results are highlighted in **bold** and underline, respectively.

Methods	ImageNet-R	ArxivQA	VizWiz-Caption	IconQA	CLEVR-Math	Flickr30k	Average
Zero-shot (Liu et al., 2023)	18.88	52.62	38.75	21.25	21.12	41.44	32.34
FT-LoRA (Hu et al., 2022)	29.33	55.30	45.51	26.13	13.07	58.07	37.90
MoE-LoRA (Chen et al., 2024)	58.43	77.57	44.83	68.90	56.73	**58.27**	60.79
Replay-LoRA	76.93	87.07	54.31	56.43	36.40	55.94	61.18
CL-MoE (Huai et al., 2025)	64.12	78.38	44.83	62.00	50.75	58.06	59.69
HiDe-LLaVA (Guo et al., 2025a)	87.62	91.12	42.68	57.62	31.00	50.41	60.08
ModalPrompt (Zeng et al., 2025)	80.50	90.62	60.13	63.50	55.75	57.09	67.93
DISCO (Guo et al., 2025b)	88.88	**94.25**	47.52	69.50	60.75	56.32	69.54
SAME (Xie et al., 2026)	**89.91**	91.40	55.33	77.51	68.85	55.43	73.07
CRAM (Ours)	87.23	92.17	**60.55**	**80.43**	**70.67**	57.35	**74.73**

Table 2: Performance on TriGap. The best and second-best results are highlighted in **bold** and underline, respectively.

Methods	PMCVQA	DocVQA	ChartQA	IconQA	InfographicVQA	ArxivQA	Roadside	ChemVQA	FloodNetVQA	CLEVR-Math	Average
Zero-shot (Liu et al., 2023)	35.40	12.68	9.36	19.27	5.06	53.77	7.40	5.30	47.41	20.37	21.60
FT-LoRA (Hu et al., 2022)	34.20	23.32	9.84	37.07	23.53	83.83	7.00	12.70	80.31	60.27	37.21
Replay-LoRA	33.70	33.95	14.00	46.67	28.97	75.57	9.40	15.90	73.81	58.80	39.08
MoE-LoRA (Chen et al., 2024)	39.03	37.49	12.44	43.43	35.17	90.90	7.93	20.70	**90.41**	**67.00**	44.45
HiDe-LLaVA (Guo et al., 2025a)	37.00	33.20	10.52	41.97	24.09	79.20	7.73	11.17	57.39	23.00	32.53
ModalPrompt (Zeng et al., 2025)	38.23	38.23	11.92	44.73	37.37	84.47	10.13	12.43	71.52	52.50	40.15
CL-MoE (Huai et al., 2025)	40.53	36.79	13.72	52.70	32.27	**93.00**	7.77	18.33	80.09	65.90	44.11
SAME (Xie et al., 2026)	41.60	43.87	17.56	64.03	**39.57**	90.46	10.83	21.77	81.09	54.50	46.53
DISCO (Guo et al., 2025b)	42.03	43.50	**18.01**	63.13	38.23	91.27	11.02	22.13	80.25	55.87	46.54
CRAM (Ours)	**43.60**	**44.33**	16.92	**66.00**	39.17	91.83	**12.83**	**24.27**	79.49	59.53	**47.79**



Resource Efficiency

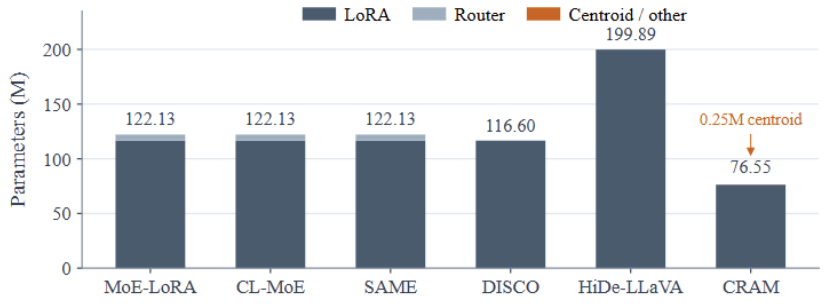


Figure 4: Parameter comparison on TriGap.

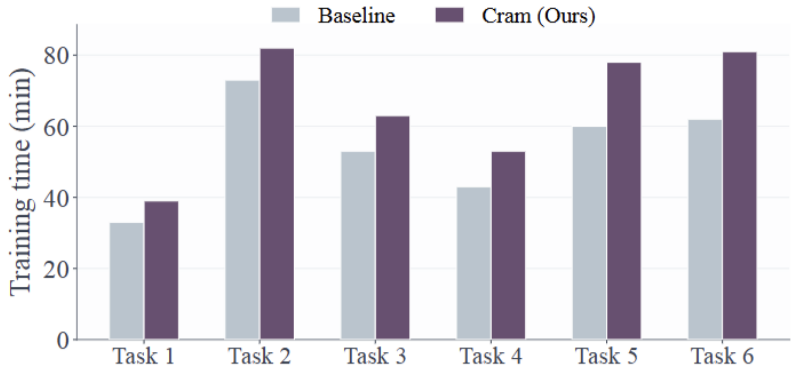


Figure 5: Training time comparison on UCIT.

Table 5: Resource efficiency comparison.

Method	Memory (MiB)	Throughput (samples/s)
SAME	28,971	3.96
MoE-LoRA	28,561	3.49
CRAM (Ours)	29,055	**4.06**

- CRAM introduces a modest training-time overhead, while memory usage remains nearly unchanged.

- It reduces trainable parameters by 37% and achieves the highest inference throughput.



Conclusion

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- Continual learning requires acquiring new capabilities without forgetting previous ones.
- CRAM isolates conflicting formats and reuses transferable visual capabilities.
- It allocates parameters only to capability gaps, achieving better performance with fewer parameters.

Code is available at :
https://github.com/LAMDA-CL/EMNLP2026-CRAM

